1 Causal Inference Objectives

Distinguishing between **experimentally-generated** and **observational data** to evaluate causal claims. Key goals include fitting regression models that adjust for **confounding variables** and applying the principle: `block what you can, randomize what you cannot` in experimental designs like A/B testing.

Relevance to Data Science This block emphasizes **"joined-up thinking"** across the data pipeline. It covers randomized studies, such as **A/B testing** for website optimization, and non-randomized studies, such as assessing drug efficacy in large healthcare databases.

2 Hypothesis Testing Review

Decision	H_0 **is True**	H_0 **is False**
Fail to Reject	Correct (TN)	Type II Error (FN / Power=1-β)
Reject H_0	Type I Error (FP / α)	Correct (TP / Power)

(H_0) represents the status quo in our case study, whereas (H_a) represents our hypothesis of interest.

The **p-value** is the probability of obtaining a test statistic as extreme as observed, assuming H_0 is true. Reject H_0 if `p-value < alpha`. The significance level α will be our tolerance to type I error: rejecting the null hypothesis when in fact is true.

Increasing Power $(1-\beta)$**:** Power is the ability to detect a real effect when it exists (reducing Type II errors). Increasing power helps you find true positives, but it doesn't stop "fake" positives from appearing due to multiple testing.

```
hypothesis.test <- function(alpha = 0.05, p_value) {
  reject <- (p_value < alpha)
  return(reject)}
```

2.1 P-Hacking & Type 1 Error Inflation

p-hacking is conducting many tests and only reporting significant results ($p < 0.05$). This inflates Type I error. For m independent tests, the probability of at least one Type I error is $1-(1-\alpha)^m$.

3 Bonferroni Correction

3.1 Family-Wise Error Rate (FWER)

Sets a stricter significance level: `alpha / m`. It ensures the **FWER** (chance of > 1 false rejection) is $< \alpha$.

```
sum(pval < (sig_alpha / (nvars * nvars))) # Number of rejected H_0
p.adjust(pval, method = "bonferroni") # Bonferroni Adjustment in R
```

3.2 Trade-offs

Bonferroni is **highly conservative**. It significantly increases the risk of **Type II errors** (missing real effects).

```
fdr_p <- p.adjust(raw_p, method = "fdr")# FDR/BH Adjustment in R
```

4 False Discovery Rate (FDR)

Controls the **expected proportion of false positives** among all rejections. The **Benjamini-Hochberg (BH)** method ranks m p-values ($p_{(1)} \leq \cdots \leq p_{(m)}$) and finds the largest k such that $p_{(k)} \leq \frac{k}{m}\alpha$.

4.1 Selecting a Correction

Bonferroni:if you prefer to be very conservative and would rather miss a real effect than report a fake one. **FDR**:if you prefer to be less conservative and want to ensure you don't miss important discoveries, even if it means a small, known percentage of your results might be false positives.

5 Simpson's Paradox

Simpson's paradox occurs when a trend appearing in aggregated data reverses when the data is separated into specific groups. This often results from a limited or poor Exploratory Data Analysis (EDA).

A real-world example is COVID-19 fatality rates: Italy appeared to have a higher overall rate than China, but when disaggregated by age, Italy actually had lower rates in every single age group.

To avoid misleading conclusions, it is extremely important to know how our data was collected. Define what variables are relevant to our study. For example, if we want to infer causality, we must determine the possible causes for our outcome of interest.

6 Confounding Variables

A variable C is a confounder of the relationship between exposure X and outcome Y if:

- It must be related to the outcome (prognosis or susceptibility).
- Its distribution must be different in the groups being compared.

Confounding is essentially a mixing of effects where the third factor independently affects the risk of the outcome.

```
simpson_data <- simulate_simpson(n=200, groups=4, r=0.4)
cor(data$X, data$Y) # May be negative
data %>% group_by(Group) %>% summarise(corr = cor(X, Y)) # Positive
```

7 Dealing with Confounding

Observational Studies: Use stratification or regression adjustment. Include confounders C as additional regressors to "block" their effect.

Experimental Studies (A/B Testing): Randomization is the "gold standard." It guarantees that treatment X is independent of all confounders, allowing for causal interpretation.(ANOVA)

8 TikTok Simulation Case Study

The "true" effect of the new ad was an 8-sec inc in dwell time. **Naive Observational**: Effect overestimated because high-dwell-city users chose the new ad. **Randomized Study**: Comparing "New" vs "Current" yielded an accurate estimate (8.22s). **ANCOVA**: Including confounders (city/age) in a randomized model increases precision.

```
# Observational with Adjustment
lm(y_obs ~ x_self_choice + city + age, data = sample_TikTok)
# Randomized Study (ANCOVA)
lm(y_exp ~ x_randomized + city + age, data = sample_TikTok)
```

Causal Inference To avoid misleading conclusions, know how data was collected. Randomization allows accurate estimation without knowing every confounder, but including known covariates (ANCOVA) makes estimations more precise.

9 Proportion Testing (Binary Y)

When comparing means of 2 groups, **2-sample t-test**

When dealing with proportions of i groups, **Z-test**

Used for Click-Through Rates (CTR). $H_0: p_1 = p_2$.

- **prop.test()**: Uses χ^2 approximation. Set `correct=FALSE` to disable Yates' continuity correction for large samples.
- **One-sided**: Use `alternative="greater"` if testing if a new variant outperforms a baseline.

```
prop.test(x = c(clicks1, clicks2), n = c(n1, n2), alternative =
      "greater", correct = FALSE)
```

9.1 Multiple Comparisons (A/B/n)

When testing > 2 groups, the **Family-Wise Error Rate (FWER)** inflates. We must adjust p-values.

- **Bonferroni**: Strict. $p_{adj} = \min(1, p \times m)$. High control of Type I error, but low power (high Type II error).
- **Holm**: Sequential, less conservative than Bonferroni. Good balance for general business/marketing use.

```
pairwise.prop.test(x = df$clicks, n = df$impressions,
       p.adjust.method = "holm")
```

10 ANOVA and Factorial Designs

Main Effect: The avg impact of one factor (Scent) ignoring others.

Interaction: When the effect of one factor changes across levels of another (e.g., Scent helps Cute lines more than Direct lines).

Visuals: Crossing or non-parallel lines in interaction plots = interaction.

```
mod <- aov(Receptivity ~ Scent * PickUp, data = d) # (Continuous Y)
interaction.plot(d$Scent, d$PickUp, d$Receptivity) # InteractionPlot
```

11 Advanced Blocking

11.1 RCBD Design

Randomized Complete Block Design: Groups units into homogeneous blocks (e.g., `YearInSchool`) to reduce "nuisance" variance.

- **Goal**: Increase precision of treatment effect estimates.
- **Rule**: Treat blocks as additive terms (`+`), not interactions (`*`), unless the block itself is of scientific interest.

```
aov(Y ~ Block + FactorA * FactorB, data = d)# Model with Blocking
```

12 Post-Hoc Reasoning

Class of experimental models refers to all possible pairwise comparisons among the levels of the main and interactions effects after the model is estimated.

Tukey HSD: Best for balanced ANOVA main effects/interactions.keeps the family-wise error rate (FWER) at or below.

PostHocTest (DescTools): Flexible for various adjustments.

```
aov_res <- aov(formula = Duration ~ Colour * Font, data = ABn_data)
tidy(TukeyHSD(aov_res, conf.level = 0.95))
PostHocTest(mod, which = "Scent:PickUp", method = "bonferroni")
```

13 Experimental Design & Strategy

Blocking guarantees that each strata or demographic category is represented equally in both treatment groups, preventing imbalances that occur more frequently in small samples.

Non-blocking: Sampling n units and randomly assigning treatments without considering strata. **Blocking**: Randomizing treatments *within* each demographic category (block). This ensures sufficient representation across categories, which is critical for small sample sizes and controlling "nuisance" variance.

13.1 Modelling Frameworks

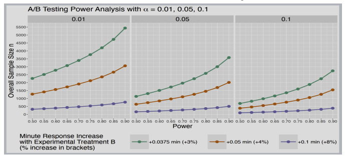
The **Raw Model** regresses response y on treatment x only, ignoring stratum effects. The **Adjusted Model** uses post-hoc inclusion of demographic covariates. The **Blocking Model** uses data from a design where treatments were randomized within strata. While all models are accurate as $n \to \infty$, blocking models remain significantly more precise.

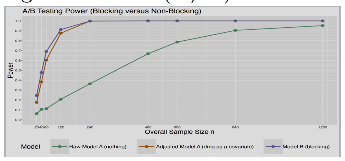
14 Monte Carlo Metrics

14.1 Performance Evaluation

Simulations track behavior via four key metrics:

- **BIAS**: Accuracy ($E[\hat{\theta}] - \theta$). Smaller is better.
- **RMSE**: Captures both bias and variance; typical error magnitude.
- **COVERAGE**: % of 95% CIs containing the true effect value.
- **POWER**: Probability of detecting a real effect ($\theta \neq 0$).





15 Statistical Power Analysis

Power $(1-\beta)$ is the probability of rejecting H_0 when H_a is true. There is a direct trade-off between significance level (α) and power. Sample size n depends on effect size (δ), variance (σ), α, and power. Stricter α (e.g., 0.01) dramatically increases the required n.

15.1 Sample Size Computation

For continuous variables, use Cohen's $d = \delta/\sigma$. Blocking can achieve acceptable power with significantly less data (e.g., 10x less in high-variance contexts).

```
# Sample Size for Two-Sample t-test
pwr.t.test(d = delta / sigma, sig.level = 0.05, power = 0.8, type =
      "two.sample", alternative = "greater")
```

16 Block Homogeneity

The Efficiency Principle Blocking works best when blocks are **homogeneous**. Stratify units so variation in Y is maximized *across* blocks and minimized *within* blocks. If within-block variation exceeds between-block variation, the benefits of blocking are negated.